Proper Statistical Tests

In statistical testing, the preferred process is a controlled experiment so that the analyst can observe the outcomes from two separate processes under identical market conditions (e.g., Wilcoxon signed rank test). While this is an appropriate technique for static strategies such as VWAP and POV algorithms, it is not an appropriate technique for those algorithms with dynamic trading rates and/or those that employ real-time adaptation tactics. Employing a controlled experiment for dynamic algorithms will likely cause the algorithms to compete with one another and will lead to decreased performance. For dynamic algorithms (e.g., implementation shortfall and ultra-aggressive algorithms) it is recommended that investors utilize the two sample non-pair approach and the Wilcoxon-Mann-Whitney ranks test.

In theory, it is appropriate to compare algorithms with static strategies (e.g., VWAP and POV) with the Wilcoxon-Mann-Whitney ranks test. However, doing so causes more difficulty with regards to robust categorization and balanced data requirements. It is recommended that algorithms with static parameters be compared via the Wilcoxon signed rank test approach.

Small Sample Size

In each of these statistical techniques it is important to have a sufficiently large data sample in order to use the normal approximation for hypothesis testing. In cases where the sample sizes are small (e.g., n and/or m small) the normal distribution may not be a reasonable approximation methodology and analysts are advised to consult statistical tables for the exact distributions of T_n and T. We recommend using at least $n > 100$ and $m > 100$ for statistically significant results.

Data Ties

It is assumed above the results are samples from a continuous distribution (i.e., statistically there will never be identical outcomes). Due to finite precision limitations, analysts may come across duplicate results, inhibiting a unique ranking scheme. In these duplicate situations, it is recommended that the data point be included in the analysis twice. In the case that algorithm "A" is the better result for one data point and algorithm "B" is the better result for the second data point, a unique ranking scheme will exist. If the tail areas of the results are relatively the same this approach should not affect the results. If the tail areas are different this may be a good indication that the data is too unreliable and further analysis is required. Analysts with strong statistical training may choose alternative ranking schemes in times of identical results.